HOMEOPATHY RESEARCH INSTITUTE

Data analysis of Australian report 2025

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Overview of report process

Systematic Search	Comprehensive search across different databases
Studies Selection	Reviewers were selected primarily RCTs; quasi-randomized, cluster, and crossover trials accepted. Non-randomised studies included only if no RCTs available .
Population prioritisation	In parallel, 20 conditions were, based on PRACI survey of homeopathy use in Australia
Outcomes prioritisation	7 outcomes domains were selected according to primary outcomes studies and outcomes measurement tools + from Core Outcome Sets, Relevant Cochrane Reviews…
Risk of biais	Cochrane RoB 2.0
Data synthesis	Quantitative synthesis (meta-analysis) & Qualitative synthesis
Conclusion	Certainty of Evidence (GRADE) and conclusion

Overview of report results

254 RCTs identified as eligible for inclusion in the review

Included	Excluded
93 studies	161 studies
78 studies	137 studies

Overview of report results

93 studies included in the quantitative analysis

- 66 RCT vs Placebo
- 13 RTC vs Inactive Control
- 14 RCT vs Active Control

78 studies included in the qualitative analysis

- 66 RCT vs Placebo
- 12 RTC vs Inactive Control (study evaluated placebo and inactive control therefore study was only included in primary comparison

Overview of report results

161 studies met the eligibility criteria but

- 21 RCT vs Active control
- 3 RCT not identified clearly the comparator

137 studies met the eligibility criteria but not conducted in prioritised population

Methodology comparison 1 : Studies selection

Item	PROSPERO V1	Full PROSPERO	Final Technical Report	Differences observed	Methodological implications	Cochrane/GRADE compliance	Comments
Condition/domain	Broad, undefined	Focused on AU conditions	Same as full		Higher heterogeneity	Not compliant (Cochrane 3.2)	
Eligible studies	RCTs	Adds cluster, crossover, quasi-RCTs	Same		Higher bias risk	Partially non-compliant (Cochrane 23.2.2)	
Participants	Any age/condition	Same	Same				
Ineligible participants	Healthy excluded	Same	Same				
Population prioritisation	Not mentioned	ICD-11 + 20 groups + umbrella group	Mention a list of 25 conditions + ICD-11 + regroup into umbrella group up to 6 condition + 20 groups	Major differences	Post-hoc bias risk	Not compliant (Cochrane 3.2)	the regroupment in umbrella condition seems to be différents between full Prospero and Final technical report
Protocol modifications	Not mentioned	Not mentioned	Queries raised during research leading a modification of the selected condition (reorganisation)	Difference whitout any specific justication	Post-hoc subgroup bias	Not compliant (Cochrane 9.6.2)	Induce less conditions
Outcome role	Not used for inclusion	Same	Same				Reviewers used primary outcome for population prioritisation
Outcome domains	Large list, max 7 critical	Same + Any effectiveness outcome anticipated to demonstrate a treatment achieves, all timepoints, all outcome measure	Same		More details		
Out of scope	Safety, PREMs excluded	Same	Same				

Methodology comparison 2 : Data analysis

Item	PROSPERO V1	Full PROSPERO	Final Technical Report	Differences observed	Methodological implications	Cochrane/GRADE compliance	Comments
Contacting authors	Not described	Authors not contacted	Same		Missing data bias	Not compliant (Cochrane 7.5)	
Approach to assess effect intervention	1- ITT + imputation 2- 2.Modified intention-to-treat analysis (i.e., an analysis that adheres to intention-to-treat principles except certain data are justifiably not included). 3- An 'as-treated' or 'per-protocol' analysis	Same	Same		Risk from imputation	Cochrane 8.13 cautions	Contradictory with later section "no imputation will be made")
Measures of effect	Treatment effect reported : RR/SMD basic	Treatment effect reported : RR/MD/SMD/HR Clinical relevance : if MCID available, the magnitude of effect (if important) will be considered on three levels: small (MD <10% of the scale), moderate (MD between 10% to 20% of the scale), or large (MD more than 20% of the scale). In the absence of an MCID or when SMD is used for different measures of an outcome, Cohen's guidance for interpreting the magnitude of the SMD will be used	Treatment effect reported : RR/MD/SMD/HR Clinical relevance : if MCID no available, the magnitude of effect (if important) will be considered on three levels: small (MD <10% of the scale), moderate (MD between 10% to 20% of the scale), or large (MD more than 20% of the scale). If effects were quantified by SMD, Cohen'sguidance for interpreting the magnitude of the SMD will be used	Changement of how to interprepret clinical relevance	Risk of threshold manipulation	Not compliant (GRADE 5.4)	
Risk of bias (study)	RoB2	Same	Same				

Methodology comparison 3 : Data analysis

Item	PROSPERO V1	Full PROSPERO	Final Technical Report	Differences observed	Methodological implications	Cochrane/GRADE compliance	Comments
Risk of bias (global)	Not mentioned	In GRADE evaluation, bias of publication (impact of missing data) will be judged based on available information, including studies not included in the review (e.g., ongoing or awaiting classification). No additional actions will be undertaken	Same		• Risk of underdetecting publication bias • No proactive search for missing data • Mixing "not included" studies with true publication bias issues • The approach is descriptive and subjective	Not fully compliant (GRADE Handbook, Ch. 8: "Assessing publication bias")	
Data transformation/ adjustement/ imputation	None	Cochrane Ch.6 calc	Only simple transformation such as SD but "no additional transformations" also stated. Or no imputation	Major differences	• Internal contradiction: no reproductible, no clear, • Post-hoc risk of bias	Not compliant (Cochrane 3.2)	
Extraction from graphs	Not mentioned	Implied no	Explicit no	Minor difference	Loss of usable data	Not optimal (Cochrane 6.5.2)	
Missing data	Not described	Cochrane Ch.6,if possible Studies with missing data will be included alongside other studies in narrative synthesis or in the forest plot for showing the sample size Missing date from studies no included in the final analysis or ongoing trial will be considered when interpreting the final grade evidence	Same		• Inflated perception of the evidence base • Increased risk of interpretative bias • Distortion of GRADE assessments (Absence of author contact increases data uncertainty) • Potential for post-hoc flexibility	Partially non-compliant (lack of transparency, misleading forest plots presentation, lack additionnal information from author)	

Methodology comparison 4 : data synthesis

Item	PROSPERO V1	Full PROSPERO	Final Technical Report	Differences observed	Methodological implications	Cochrane/GRADE compliance	Comments
Quantitative synthesis	RE model, stratified, Forest plot stratification: Individualised vs non-individualised	Same	No specified the forest plots stratification	Final changes stratification	Loss of clinically relevant subgroup + mixed with qualitative synthesis Lakc of transparency	Not compliant (Cochrane 10.11)	
Qualitative synthesis	Narrative summary	Narrative summary, tables, forest plots (no specified stratification,Vote-counting)	Forest plot stratification: outcomes measures tools	Final changes stratification	mixed with qualitative synthesis, induce perception of the evidence base	Not compliant (Cochrane 12.2.1)	

Population prioritisation

The NHMRC report used a structured, multi-step procedure to select the 20 clinical conditions, based on PRACI survey results and equivalent Australian survey data.

- Inconsistency in preliminary condition number (25 conditions??)
- Lack of Transparency in Transitions and Grouping, loosing some conditions
- Process Not Conducted Blindly
- Omission of Clinically Relevant Conditions: These exclusions were not justified, despite their clinical relevance and presence in eligible studies.

Table A-10 Final revised list of priority conditions for homeopathy

NTWC Rank	REVISED PRIORITY CONDITIONS FOR DATA SYNTHESIS
1	Anxiety
2	Atopic conditions (allergies, hay fever, eczema)
3	Headache/migraine
4	Digestive disorders (infantile colic)
5	Digestive disorders (infantile diarrhea)
6	Irritable bowel syndrome
7	Recurrent infections (childhood otitis media)
8	Recurrent infections (URTI)
9	Insomnia/sleep disorders
10	Fatigue conditions (e.g., post viral fatigue, ME/CFS)
11	Depressive/mood disorders
12	Fibromyalgia
13	Arthritis
14	Asthma (prevention of)
15	Recurrent infections (UTI)
16	Psoriasis
17	ADHD/Autism/Learning difficulties
18	Menopausal symptoms
19	Menstrual disorders
20	Back pain/ neck pain
Not included (moved or removed from initial ranking) ^	Pregnancy/childbirth conditions[a]
	Inflammatory bowel (e.g. Crohns, colitis)[b]
	Recurrent infections (warts)[c]
	Recurrent infections (vulvovaginal candidiasis)[c]

^ It was intended that a maximum of 20 priority populations would be considered in the evidence synthesis (estimated maximum 100 RCTs).
a. Pregnancy as a condition was ranked lower priority than menopause and menstrual disorders because it was considered that studies in pregnancy/childbirth could be included within other higher priority conditions based on the reported outcome or study focus (e.g., recurrent infections [UTI], insomnia etc.).
b. No studies found in this condition, therefore not included in the total count.
c. Removed to reduce the volume of evidence reviewed in the umbrella group.

Outcomes prioritisation

For each prioritised population, the evidence reviewers compiled a list of all supplemented with data from Core Outcome Sets (e.g. COMET, ICHOM), relevant Cochrane reviews, and expert input from NTREAP. Each condition retained up to seven critical or important outcomes, which were then presented in the Summary of Findings (SoF) tables

- Narrowing of the Evidence Base: Outcome prioritisation was conducted only for the 20 prioritised populations
- Loss of Clinical Context
- Partial Blinding
- Opaqueness of Scoring Process

Risk of bias : RoB2

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- Evaluation on the primary outcome
- Unclear criteria for selecting the primary outcomes (in studies with several primary outcomes)
- Inconsistency with Cochrane standards
- Opaque reporting and limited transparency

Data analysis

The review's process systematically excluded a large volume of evidence through overly strict rules and a failure to follow best practices.

- Refusal to pursue missing data
- Rejection of standard data transformation methods
- Inconsistency between methodological framework and its actual implementation
- Lack of consistency and transparency

Data synthesis

Quantitative synthesis (meta-analysis):

- Only for the studies compare homeopathic medicine with placebo or negative control.
- Effect estimate across the studies will be combine using random effect model (RevMan 5.4?)
- Forest plot stratified by the type of homeopathic medicine (individualised or not).
- Statistical heterogeneity using the Chi^2 test and quantifying heterogeneity using the I^2 statistic

Qualitative synthesis:

- Narrative synthesis of the results for each condition with the risk of bias
- Summary of findings grouped by comparator and outcome domain with forest plots when is possible grouped by risk of bias (without pooled estimates).
- Description of an overall effect across multiple studies for each outcome within the GRADE summary off indings tables with the magnitude, range and distribution of observed effects across the studies using a simple vote count based on direction of

Data Analysis

Forest plot:

1. The total participant counts include studies with non-analysable data, which creates an illusion of greater statistical power and internal inconsistencies with the weighted effects. (Cochrane sections 10.3.1 and 10.10.3.)

2. There are two forest plots (MD and SMD) for outcomes with only one usable study. SMD is designed for pooling studies with different measurement scales, so it's not applicable here (Cochrane 10.9.2.)

3. Summary diamonds and overall effects are shown when no meta-analysis was actually performed. This makes it impossible for readers to distinguish between qualitative synthesis and quantitative pooling.(Cochrane 10.10.3.) The forest plots are presented in a way that misleads readers about the actual evidence strength and lacks transparency about which outcomes had proper meta-analyses versus qualitative descriptions

Major Failures 1

1. Lack of transparency

- Difficult to reproduce how outcomes were rated or prioritised across the report.
- Impossible to track or compare the studies that were excluded from the data synthesis.
- Lack of visibility between qualitative assessments and quantitative synthesis (forest plots).
- Overall, the transparency of the analytical workflow is insufficient to allow replication.

2. Internal inconsistencies

- The methodological process appears to shift depending on the section, making it difficult to understand the logic of the review.
- Several procedural steps remain unclear or undocumented.
- The method seems to evolve throughout the analysis, without justification or documentation.
- These inconsistencies reduce confidence in the robustness of the process.

3. Selection bias and limited reproducibility

- Study selection is sometimes arbitrary or poorly justified, leading to a drastic reduction in included evidence.
- Lack of reproducibility in the selection strategy.
- Exclusion criteria appear inconsistently applied, which may eliminate clinically relevant evidence.
- Resulting evidence base may be non-representative of the available clinical landscape.

Major Failures 2

4. Inconsistency with guidelines (GRADE, Cochrane, RoB2)

- Unclear or incorrect use of the RoB2 framework.
- Mixing qualitative forest plot results with non-estimable or missing data.
- Inclusion of participants whose outcomes were not evaluated or not estimable into total sample sizes.
- This artificially inflates denominators and can mislead the reader.
- Several methodological choices contradict explicit guidance from GRADE and Cochrane regarding evidence synthesis and missing data management.

5. Impact on interpretation

- These limitations collectively lead to a distorted perception of the evidence base.
- Conclusions may be driven more by methodological artefacts than by actual data.

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Thank you for your attention!

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